

CREATIVELY GREEN

An Efficient and Eclectic Design for a Creative and Active Family

BY ELKA KARL

When Betsy Teter and John Lane, a writer and a college professor in South Carolina, decided they wanted to build a new house, they knew what they wanted. In fact, they were quite specific about their aesthetic, describing a home that was a cross between Frank Lloyd Wright and Henry David Thoreau. “In other words, we wanted self-sufficiency and efficiency with creative design,” says Teter. “We basically drew the rooms—and wide-open floor plan—on a piece of paper and they took it from there.”

The architects, led by designer/project manager/green consultant Bob Bourguignon of Sustainable Architecture in Moore, South Carolina, created a design for the house that is reminiscent of a ship under sail, with the idea that the house’s rigid geometry, cable rails, and great wall would complement the man-made environment and contrast with the natural forest surrounding it. “It scared me at first—the big 30 ft wall sticking out of the back and front,” admits Teter. “But we’re both creative people, so we just said, ‘Go for it.’”

The Teter-Lanes, along with their two teenage boys, are an active, community-oriented family. They enjoy backpacking and kayaking, and are also involved in local river cleanup and preservation projects. They are also advocates of smart growth and sustainability. For these reasons, they worked to ensure that their new home was energy efficient, used green materials, had plenty of light and views, and had the minimum impact on their forested site.



BOB BOURGUIGNON

The design for the great wall provides dramatic contrast with the heavily wooded property site.

Site Strategies

The Teter-Lanes used the LEED 2.0 Rating System as a guide to planning construction. They also took a holistic design approach. When choosing the site, the Teter-Lanes looked for a wooded parcel of land that was close to Lawson’s Fork, the river where they kayak and also participate in cleanup projects. Finding a site that was a reasonable distance to the closest town was also important to them. The site they chose is less than 5 miles from downtown Spartanburg, South Carolina, which means that the Teter-Lanes can minimize energy spent on transportation.

After they chose their 1 acre site, the Teter-Lanes decided that they wanted to preserve the tree canopy whenever possible. With this in mind, they limited cutting of the forest to a 10 ft strip surrounding the residence and a small area for the septic field. Where large trees were located within that area, builders worked around them. Using this strategy, 60% of the forested site was left undisturbed. Most home sites are nearly clear-cut during construction. (LEED awards one point if 25% or more of the site is undisturbed, based on a home-sized lot of up to 1 acre.) The remainder was planted with native shrubs and perennials, with no turf, eliminating the need for mowing and irrigation. There was no need to cre-



BETSY TETER



BETSY TETER

(left) The Teter-Lanes wanted an open floor plan for their home. (right) The south-facing windows reduce the need for artificial lighting.

ate a new driveway, either, since the Teter-Lanes could access both their home and the river from an existing dirt road.

Since the Teter-Lanes are active kayakers, they decided to forgo a garage and instead dedicated one room of the house to kayak storage. They also considered the needs of the family as a whole. Because their boys are interested in music (and their parents are interested in keeping their sanity), they designed the house so the boys would have their own floor, where they could play their guitars without disturbing their parents. Finally, the home had to fit within a tight budget.

Although the Teter-Lanes were only vaguely familiar with home performance issues, they and their architects did substantial research on the building performance and green building concepts to learn how to apply these concepts to their new home.

During and after construction, many strategies were employed to make sure that the site remained relatively undisturbed. During construction, silt fences and check dams were used to keep eroded material out of the river and the surrounding floodplain. Strategically placed stones collected rainwater in miniponds around the site, allowing it to

soak into the soil rather than running off into the river. After construction was finished, the Teter-Lanes also created a dry streambed, using 8 tons of river rocks. The streambed runs three-quarters of the way around the house, funneling all the water from the driveway and the roof into a natural bog area between the house and the river. “John has named the dry bed ‘the Little Betsy,’” says Teter, “and it really roars when it rains. Otherwise, all that water coming down the driveway would have shot straight down the hill into the river. Now it never makes it to the river.”

Efficient and Green

The high-efficiency building shell—6-inch wood stud exterior walls with self-adhered, blown-in recycled shredded newspaper cellulose insulation—provides up to R-22. The 9 1/2 inch truss joists—rafters with self-adhered cellulose insulation—provide up to R-37. All framing lumber, as well as and including wood truss floors, laminated veneer lumber, and parallam beams, is engineered wood and oriented strand board sheathing.

The massive block fireplace helps to reduce the need for heating and also to

reduce heat fluctuations. The great wall that runs through the home was originally intended to be a thermal mass with two wythes (a wythe is a continuous vertical section of a masonry wall, composed of one layer of block or brick). However, budget constraints obliged the Teter-Lanes to reduce the wall to one wythe, which made the thermal mass benefits of the wall negligible; due to this change, the only masonry that acts as a thermal mass is the chimney.

Energy Star windows from Eagle Windows and Doors were used throughout the house (with the exception of the entry, which has a custom door in an Eagle frame). Large windows with operable sashes on the south elevation and smaller high windows on the north elevation (all spaces one room deep) allow flow-through ventilation. An exhaust fan located high on the wall in the 18 ft tall living room exhausts hot air from the house, replacing it with cooler fresh air in the summer. The strategic opening of windows, and the combination of the fan and windows, can keep the house cool much of the year, greatly reducing the need for air conditioning.

Large south-facing windows combined with roof overhangs, wall exten-

sions, overhanging lattices, and the close proximity of the forest make the interiors bright without excessive summer solar gain. During the winter the sun angle and the lack of foliage allow for more solar gain. The windows frame pleasing views of the forest that surrounds the house. Overhangs were used to help with cooling and heating as well. The eaves shade the windows during the summer when the sun is high and let in the sun (and solar gain) during the winter. The great wall also helps to shade the living space from the afternoon sun. The home is daylit with large areas of shaded south facing glass, which eliminates the need for most lighting during the day.

One of the decisions that Teter took responsibility for was the installation of the roofing material, which is high albedo galvalume metal, with a reflectivity of .80 new and .65 aged. "Our contractor didn't want to mess with it, so he quoted on shingles," she says. "Right before it was to go on, I took control and went and found a subcontractor and had it put on. I liked the look, plus I liked the idea that it would help the utility bills."

Inside the house, efficient fixtures and appliances were installed to reduce the Teter-Lanes' use of resources. Water use was reduced through the installation of low-flow fixtures, and newer refrigerants were used in the house's HVAC system. On average, the power bill for the 2,800 ft² house is \$120 monthly, while the gas bill averages \$66 a month. Most of the year the gas bill is significantly lower, but it rises in the winter, due to increased heating costs.

Frustrations and Compensations

In any complicated, daunting project, there are bound to be setbacks, and the Teter-Lanes' house was no exception. Not everything went as smoothly as they would have liked. Though the interior floors are made of rapidly renewable bamboo, they have not held up in the last two years as well as the Teter-Lanes had hoped. "We had a hard time finding subcontractors to do some of the things we wanted done," says Teter. "I believe it



The Teter-Lanes limited the cutting of trees to a 10 ft strip surrounding their home, enabling them to preserve 60% of the forest.

was the first bamboo floor in our town. The finish is chipping off and we are going to have to redo it soon." It is as much the same story with the concrete countertops. "The guy doing the concrete countertops had never done them before. It took many tries, and the quality is a little inconsistent."

The Teter-Lanes also found it difficult to be as green with their building process as they wanted to be, even when they tried to be diligent. Although their construction waste was supposedly being sorted and recycled by a LEED-certified recycler, they later found out that the contractor, who left in the middle of the job, hadn't actually recycled any of the waste. However, the installation of green materials—such as low-VOC water-based paints on all interior surfaces, polished C.M.U. pellant walls used as a structural and finish material, and cellulose insulation—offset these setbacks and ensured that the Teter-Lanes' efforts weren't all for naught.

The Teter-Lanes moved into the house in May 2003. Since then, it has more than met their expectations. John Lane is currently writing a book about the house-building experience, and both Teter and Lane find that the house is conducive to their writing. When

asked which is her favorite element of the house, Teter doesn't have to think twice to come up with her answer. "The windows! The whole south-facing side is windows. The light is fabulous," she raves. "Also, we are both writers and we have bookshelves enough for 5,000-plus volumes." When asked about the design of the house itself, Teter admits that she wasn't always as enthusiastic as she is now about the slightly unorthodox design. The great jutting wall that defines the shape of the house was originally going to be stone, though that plan was nixed after the Teter-Lanes received an estimate of \$100,000 for the wall alone. "That's when we went back to concrete block," says Teter. "The day the concrete was delivered to the site, I had another low moment, thinking we had made a terrible mistake. It was just so different than any of the architecture in this very conservative area. Then John found a book about the architecture of Clark and Menefee—they use lots of concrete block—and I settled down. Now I wouldn't have it any other way."

Elka Karl is an associate editor at Home Energy.

